1 — Problem setup

$\dot{x} = -0.1\,yz, \quad \dot{y} = 1.6\,xz, \quad \dot{z} = -0.75\,xy, \qquad u_0 = (0.5,\ 0.75,\ 1.0)$

$t \in [0,\ 13.26446]$ · 27 windows · 1,536 points/window · float64

REPEAT FOR EACH WINDOW $k = 1 \ldots 27$

2 — PINN for window k

- **Input** — t
- **Hidden** — 4×60 tanh
- **Output** — $N_k(t) \in \mathbb{R}^3$
- **Hard IC** — $\hat{u}_k = u_{0,k} + (t - t_k)\,N_k(t)$

3 — ODE residual

$r_k = d\hat{u}_k/dt - f(\hat{u}_k)$, derivatives by autodiff

4 — Causal residual loss

$\mathcal{L}_i = \mathrm{mean}(r_i^2), \quad w_i = \exp\left(-\varepsilon \sum_{j<i} \mathcal{L}_j\right), \quad \mathcal{L} = \mathrm{mean}(w_i \mathcal{L}_i)$

$\varepsilon:\ 0.01 \to 0.1 \to 1 \to 10$

5 — Adam optimization

learning rate × 0.9 every 1,000 steps

Stage complete?

$\min w > 0.99$ or 4,000 steps

NO

YES

All ε values used?

NO

Next ε

keep weights

next ε

YES

6 — L-BFGS

≤ 1,000 iterations

unweighted residual MSE

7 — Save window

checkpoint · history · prediction

Last window?

$k = 27$

NO

Next window

$u_{0,k+1} = \hat{u}_k(t_{k+1})$

copy weights, $k \leftarrow k + 1$

next window

YES

8 — Reference evaluation

DOP853 · 13,265 points · rtol 10^{-10} · atol 10^{-12}

used only for evaluation, never in training

9 — Evaluation outputs

MAE · RMSE · max error · residual MSE · invariant drift · training cost